

Public Safety Alerts Aggregator & Explorer

Course: Software Development 1-ITC303 / ITC306
Assessment item 1-Group Project : Project Charter and Plan
Due Date: March 30, 2026

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Project Details

Field	Your entry
Project ID / Project Name	2026-16 – Public Safety Alerts Aggregator & Explorer
Client / Sponsor	Jason Howarth
Team name	Critical Signal
Team leader	Ettore Vescio
Deputy team leader	Thomas Smith
Team members	Ettore Vescio, Christian Finocchiaro, Thomas Smith, Pradip Pandey
Submission date	March 30, 2026
Version	0.1

1. Project Charter

1.1 Problem / Opportunity Statement

Users currently must visit multiple government and emergency websites to access public safety alerts (e.g., weather warnings, bushfires, floods, road incidents). Alerts are published in different formats, making it hard to quickly identify relevant hazards. During emergencies, this fragmentation can delay decisions. A unified alert aggregation system would consolidate trusted sources in one interface, improving awareness and enabling faster safety decisions.

1.2 Objectives and Success Criteria

Objective	Success measure (how you will know it is achieved)
Aggregate public safety alerts from multiple sources	System collects alerts from at least three public sources (e.g., BoM, NSW RFS)
Store alerts in a structured database	Alerts stored in MySQL or equivalent with fields for category, region, and time
Provide a web interface for viewing and filtering alerts	Users filter alerts by category, region, and status
Demonstrate end-to-end data pipeline	System performs ingestion → normalisation → storage → API → web interface
Provide clear alert detail pages	Users see issuing agency, time, region, description, and official source link

1.3 Stakeholders and Roles

Stakeholder	Role (care/decide/use/approve)	Responsibilities / interests	Contact / channel
Jason Howarth	Project Sponsor / Client	Defines requirements, reviews progress, provides feedback	e: jhowarth@csu.edu.au ph: 0265829453
Critical Signal team	Development Team	Design and implement system features	Discord / email
CSU Students & Staff	End Users	Access alerts relevant to their region	Feedback via email, survey, or verbally

Government Emergency Alert Providers (BoM, NSW RFS, SES)	External Data Providers	Provide original public safety alerts	Public feeds and documentation
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1.4 Scope Boundaries (Current)

In scope	Out of scope
Data collection via RSS feeds and public APIs	Real-time push notifications via messaging platforms
Web UI for searching, filtering, and viewing alerts	Mobile app development
Data ingestion and normalisation	Integration with private or restricted data sources
Back-end API for alert queries	Acting as an official alert authority
Three genuine alert sources	Advanced geospatial modelling beyond feed data
Basic ingestion logging and monitoring	Machine learning or predictive hazard analytics

1.5 Constraints and Assumptions

Item	Type (Constraint / Assumption)	Implication for approach / feasibility
Selected public feeds remain active	Assumption	Alternative sources needed if a feed becomes unavailable
Focus on Australian public safety feeds (BoM, NSW RFS)	Assumption	Data limited to one state reduces collection scope (NSW)
No budget	Constraint	Limited to free services/ resources
System depends on third-party public feeds	Constraint	API changes may affect ingestion
Public feed rate limits	Constraint	System must schedule requests and apply rate limiting

1.6 Solution Context

Software Project Product Vision

For: CSU staff, students, teachers, and regional community members needing quick access to public safety alerts

The solution: Web-based alert aggregation system collecting multiple public alerts, storing in a structured database, with a searchable interface

Provides: Single, reliable location to view and understand current safety alerts

1.7 High-Level Deliverables

Deliverable	Description	Evidence of completion (what you will show)
Prototype Demonstration	Web system collecting alerts from multiple sources into MySQL	Live demo and GitHub repo
Technical documentation	Architecture and design docs	GitHub documentation
Alert ingestion pipeline	Collect, normalise, and store alerts from multiple feeds	Ingestion logs and database records
Back-end API	Retrieve and filter data from back-end to front-end	Live API data retrieval

Web Alert Explorer UI	Browse and filter alerts via web interface	UI demo with filtering and detail view
Database Schema	Structured design supporting alert storage and queries	ER diagram and schema documentation

1.8 Definition of Success

Success is achieved when users can access a working web prototype aggregating multiple public alerts, storing them in a structured database, and filtering/searching via a clear web interface.

2. Project Plan

2.1 Main Milestones (Phase Gates)

Milestone / phase gate	Target week/date	Deliverable/ decision	Owner	Success evidence
Project Charter & Plan Submission	30/03/2026	Finalised project charter and plan	Ettore Vescio	Submitted on time and meets assessment requirements
Requirements & Scope Baseline Approved	06/04/2026	Confirmed requirements, scope boundaries, and selected data sources	Team	Documented requirements and agreed system scope
Architecture & Data Model Approved	20/04/2026	System architecture diagram, database schema, and API design	Team	Reviewed and agreed design artefacts
Data Ingestion & Storage Operational	27/04/2026	Automated ingestion pipeline storing alerts from multiple sources	Team	Database populated with alerts and ingestion logs verified
Integrated Prototype Functional	15/05/2026	End-to-end working system (ingestion → database → API → UI)	Team	Demonstration of working system with filtering and alert display
Final Demo & Presentation	01/06/2026	Final system demonstration and presentation	Team	Successful demonstration meeting project objectives

2.2 Work Breakdown Summary (Major Work Streams)

Work stream	Description	Owner	Notes
Project Management	Planning, documentation, risk management, and stakeholder communication	Ettore Vescio	Includes meetings and charter

Data Ingestion	Collect alerts from RSS feeds/APIs and handle scheduling and logging	Thomas Smith	Core pipeline component
Data Processing & Storage	Normalise alerts and store in MySQL	Ettore Vescio	Includes deduplication and update handling
Back-end API	Provide endpoints for querying alerts and triggering subscriptions	Thomas Smith	Supports frontend filtering
Front-end Web Application	UI for browsing, filtering, and viewing alert details	Christian Finocchiaro	React-based interface
Testing & Quality Assurance	System testing, debugging, and validation of ingestion and UI	Pradip Pandey	Includes logging verification

2.3 Schedule Snapshot (Key Dates Only)

Week	Key date	Major deadline / internal milestone	Notes
Week 5	30/03/2026	Assessment 1 Submission	Project Charter & Plan due
Week 6	06/04/2026	Requirements & Scope Baseline	Finalise system scope, data sources, and requirements
Week 8–9	20/04/2026	Architecture & Design Complete	Architecture, schema, and API design finalised
Week 9–10	27/04/2026	Ingestion Pipeline Operational	Alerts successfully flowing into database
Week 11	15/05/2026	Assessment 2 Submission	Working end-to-end prototype
Week 14	01/06/2026	Final Demo	Presentation and system demonstration

2.4 Top 5 Risks and Mitigations

Risk	Likelihood	Impact	Mitigation action	Owner
External data source downtime or unreliability	Medium	High	Display last successful refresh time and fallback to cached data when sources are unavailable.	Pradip Pandey
Data Schema Changes by External Sources	Medium	High	Use adaptable parsers; robust error handling; log unrecognized formats.	Ettore Vescio
Data Quality and Inconsistent	High	High	Use robust normalisation and	Thomas Smith

Formats from Sources			fallback handling for inconsistent data	
Mapping Implementation Complexity	Medium	Medium	Start with simple region text filtering before attempting polygon mapping.	Christian Finocchiaro
Team Capability/ Skills Gaps	Medium	High	Opt for familiar stack (e.g., Python/Node) and hold pair programming sessions.	Ettore Vescio

2.5 Communication Cadence

Meeting / touchpoint	Purpose	Attendees	Frequency	Artefact/ output	Tool/link
Stand-up meetings	Track progress, plan tasks, and resolve blockers	Entire team	Weekly	Meeting minutes and action items	Discord
Client Progress Update	Maintain stakeholder visibility of progress	Ettore Vescio	Fortnightly	Short email progress summary	Email
Client Clarification / Review	Resolve uncertainties and validate key decisions	Ettore Vescio, Sponsor	Ad hoc (as required)	Email responses or meeting notes	Email / Zoom
Code Review	Ensure code quality and consistency	Team members	Weekly	Review comments and approvals	GitHub

2.6 Change Control Approach

Any proposed changes to scope, requirements, or deliverables will be recorded in the project issue tracker to ensure traceability. The impact on timeline and resources will be assessed by the Team Leader, and all significant changes must be reviewed and approved by the project sponsor before implementation.

3. Workspace and Collaboration Evidence

Workspace Evidence Links

Evidence item	Link / location	Screenshot ref (if used)	Notes
First Meeting Minutes	GitHub	See Appendix A	Notes from first stand-up including decisions, action items, and responsibilities.
Second Meeting Minutes	GitHub	Appendix B	Notes from second stand-up, including decisions, action items, and responsibilities.
Third Meeting Minutes	GitHub	See Appendix C	Notes from third stand-up, including decisions, action items, and responsibilities.
Fourth Meeting Minutes	GitHub	See Appendix D	Notes from fourth stand-up, including decisions, action items, and responsibilities.
Project Management Board Setup (Miro)	https://miro.com/app/board/uXjVG1LxTxY=?share_link_id=849242558963	See Appendix Figure 1	Initial board setup with project sections, workflow columns, and task cards.
Onboarding / Team Roles	https://miro.com/app/board/uXjVG1LxTxY=?share_link_id=849242558963	See Appendix Figure 2	Team members added, roles assigned, and responsibilities documented on board.
Repository / Artefact Space (GitHub)	https://github.com/Azmith47/public-safety-alerts-aggregator.git	See Appendix Figure 3	Repository created with README, initial branches, and folder structure.
First Tasks Created	https://miro.com/app/board/uXjVG1LxTxY=?share_link_id=849242558963	See Appendix Figure 4	Initial tasks for data ingestion, UI design, and backend API setup logged.
Communication / Chat Evidence	https://discord.gg/TcwBP3zZ Discord Server	See Appendix Figure 5	Discord discussion threads showing progress updates, and Q&A.

Appendix

-.1 meeting-minutes/meeting-01.md

A Meeting 1 – Project Kick-off

Date: Sunday, 8 March 2026 **Time:** 7:30 PM **Location:** Online

Chair: Ettore Vescio

Minutes recorded by: Christian Finocchiaro **Attendees:**

- Ettore Vescio
- Christian Finocchiaro
- Thomas Smith
- Pradip Pandey

A.1 Meeting Objective:

Establish the project team, review the project brief, identify initial data sources, discuss the technical approach, and assign initial setup tasks.

A.2 1. Team Introductions

Each team member introduced their background and relevant technical skills.

- **Ettore (Tory)** – Embedded software engineer at NSW Police with ~8 years of R&D experience in software and electronic systems. Skills include Python, JavaScript, SQL, HTML, and Java across front-end and back-end development.
- **Pradip (Patrick)** – Final year Cybersecurity major working in logistics. Skills include Python, JavaScript, HTML/CSS, networking, and firewall configuration.
- **Tom** – Final year Cybersecurity major with a Software Development minor. Experience with JavaScript, TypeScript, Node.js, React, Next.js, and Java. Interested in full-stack development.
- **Christian** – Final year Web Development major with experience in HTML, CSS, JavaScript, Java, and C#. Background in IT support.

The team agreed to aim for a **High Distinction** grade for the project.

A.3 2. Team Leadership

Ettore (Tory) was nominated and unanimously selected as **Team Leader**.

A.4 3. Project Overview

The team discussed the project **Public Safety Alerts Aggregator & Explorer**, which will:

- Collect public safety alerts from open RSS feeds and APIs.
- Store alert data in a MySQL database.
- Provide a web-based interface to explore alerts by region and category.
- Display alerts visually on a map.

Potential alert types include bushfires, severe weather warnings, and road incidents.

A.5 4. Data Sources (Initial)

The following data sources were identified:

- Bureau of Meteorology RSS – <https://www.bom.gov.au/rss/>
- NSW Rural Fire Service RSS – <https://www.rfs.nsw.gov.au/news-and-media/stay-up-to-date/feeds>
- NSW Live Traffic Hazards API – <https://opendata.transport.nsw.gov.au/dataset/live-traffic-hazards>

The team agreed to initially integrate **BoM and RFS feeds**, with additional sources added if time permits.

A.6 5. Technical Discussion

A.6.1 Data Collection

- RSS feeds will be used for BoM and RFS alerts.
- APIs or permitted scraping may be used for additional sources.

A.6.2 Front-End

- Web-based interface.
- **React** suggested as the preferred framework (final decision pending).

A.6.3 Back-End

- Backend language under consideration:
 - Python
 - Node.js

A.6.4 Notifications

- Potential alert notifications via **Discord API** were discussed.
 - Need to confirm whether notifications should be in-app or external (e.g., email or SMS).
-

A.7 6. Preliminary Role Allocation

Team Member	Role
Ettore	Team Leader, database design, full-stack development
Tom	Full-stack development
Christian	Front-end development
Pradip	Security testing, monitoring, and general development support

A.8 7. Key Decisions Summary

- Ettore confirmed as Team Leader.
 - Initial alert sources will be Bureau of Meteorology (BoM) and NSW Rural Fire Service (RFS) RSS feeds.
 - React selected as the preferred front-end framework (pending confirmation).
 - Backend technology decision deferred to next meeting.
 - GitHub repository and Miro project board established.
-

A.9 8. Open Questions

The following questions will be clarified with the project supervisor:

- Are external notifications (Discord/email/SMS) acceptable?
 - Is a **user login system** required?
 - Confirmation of the **final project due date** and milestones.
-

A.10 9. Action Items

Task	Owner	Due Date
Create GitHub repository and add team members	Ettore	Completed
Create project board (Miro)	Ettore	Completed
Research alert data sources and APIs	All	12 March 2026
Confirm requirements with supervisor	Ettore	12 March 2026

A.11 10. Next Meeting

Date: Sunday, 15 March 2026 **Time:** 7:30 PM

Minutes stored in:

GitHub → /docs/meeting-minutes/

B Meeting 2 – Project Charter & Plan Draft

B.1 Date & Attendees

Date: Sunday, 15 March, 2026 **Time:** 7:30 PM **Location:** Online

Chair: Ettore Vescio **Minutes recorded by:** Pradip Pandey **Attendees:**

- Ettore Vescio (Team Leader)
- Christian Finocchiario
- Thomas Smith
- Pradip Pandey

B.2 Agenda

1. Review project brief
 2. Discuss project charter sections
 3. Set up collaboration tools
 4. Identify initial data sources
 5. Plan next steps
-

B.3 Discussion

B.3.1 Collaboration Tools

The team confirmed the following collaboration tools:

- **GitHub** – source code repository and version control
- **Miro** – project planning board and task tracking
- **Discord** – team communication
- **Google Docs / Typst repository** – documentation drafting

The Miro board will be used to:

- Track tasks and responsibilities
- Visualise project workflow
- Monitor project progress

Each team member is responsible for updating their assigned tasks.

B.3.2 Project Brief Review

The team reviewed the client brief and identified the key system requirements:

- Aggregating alerts from multiple public safety sources
- Normalising alert data into a consistent format
- Storing alerts in a structured database
- Providing a web interface for search and filtering
- Designing the system so it can be expanded in Session 2

B.3.3 Project Planning

The team began drafting the **Project Charter and Project Plan**, including:

- Problem / opportunity statement
- Project objectives and success criteria
- Stakeholder identification
- Scope boundaries
- Constraints and assumptions
- High-level deliverables
- Definition of project success

B.4 Decisions Made

- Team name confirmed as **Critical Signal**
- **Miro** selected as the project planning and task tracking platform
- Initial data sources will include:
 - Bureau of Meteorology (BoM)
 - NSW Rural Fire Service (RFS)
- Current team role assignments will remain unchanged
- Remaining sections of the Project Plan will be completed in the next meeting

B.4.1 Technical Direction (Preliminary)

The team discussed the likely system architecture:

- **Data ingestion layer** to collect RSS feeds and APIs
- **Normalisation process** to standardise alert formats
- **MySQL database** for persistent storage
- **Back-end API** to expose alert data
- **React-based web interface** for users

Final technology decisions (Python vs Node backend) will be confirmed after further research.

B.5 Open Questions

The following items will be clarified with the project sponsor:

- What should the system do if an alert source becomes temporarily unavailable?
 - Does the **non-software project context section** apply to this project?
-

B.6 Action Items

Task	Owner	Due
Contact project sponsor regarding open questions	Ettore	Before next meeting
Complete remaining sections of the project plan	Team	Next meeting
Finalise charter draft for submission review	Team	Before deadline

B.7 Next Meeting

Date: Sunday, March 22, 2026

Time: 7:30 PM

Minutes stored in:

GitHub → /docs/meeting-minutes/

C Meeting 3 – Project Charter & Plan Draft

Date: Sunday, 22 March, 2026 **Time:** 7:30 PM **Location:** Online

Chair: Ettore Vescio **Minutes recorded by:** Ettore Vescio **Attendees:**

- Ettore Vescio (Team Leader)
 - Christian Finocchiario
 - Thomas Smith
 - Pradip Pandey
-

C.1 Agenda

- Client Requirements Review
- Discuss project charter sections
- Draft Project Plan (Section 2)
- Plan next steps

C.2 Discussion

C.2.1 Client Requirements Review

- The team reviewed the client requirements and confirmed the following key requirements:
- Alerts must include both active and historical data
- Users must be able to subscribe via filters (region, category, severity)
- System must handle unavailable data sources and show last refresh time
- Prototype should include 3 data sources
- System should focus on NSW, with future scalability
- Alert updates should update existing records, not create duplicates

C.2.2 Project Planning

The team continued drafting the Project Charter and Project Plan. Progress includes:

- Drafted Project Charter sections (Problem Statement, Objectives, Stakeholders, Scope, Constraints)
 - Began drafting Project Plan sections:
 - Main Milestones
 - Work Breakdown Summary
 - Schedule Snapshot
 - Risks and Mitigations
 - Communication Cadence
 - Change Control Approach
-

C.3 Decisions Made

- Selected initial data sources:
 - Bureau of Meteorology (BoM)
 - NSW Rural Fire Service (RFS)
 - State Emergency services (SES)
- Confirmed Node.js (backend) and React (frontend) as the technology stack
- Current team roles remain unchanged
- Project Plan will be finalised in the next meeting

C.3.1 Technical Direction (Preliminary)

The team agreed on a preliminary system architecture:

- Data ingestion layer (RSS/API collection)

- Normalisation process for standardising alert formats
 - MySQL database for persistent storage
 - Back-end API (Node.js)
 - Front-end web interface (React)
-

C.4 Open Questions

The following items will be clarified with the project sponsor:

- Clarify difference between Milestones (2.1) and Schedule Snapshot (2.3)?
 - Confirm how far the project schedule should extend (Session 1 only vs full project timeline)
 - Request feedback from the client on the current draft
 - Confirm availability of the Assignment 2 template
-

C.5 Action Items

Task	Owner	Due
Finalise Project Plan sections	All	Before next meeting
Refine milestones and schedule	Ettore	Before next meeting
Update Miro board with current tasks	All	Ongoing

C.6 Next Meeting

Date: Sunday, March 29, 2026

Time: 7:30 PM

Minutes stored in:

GitHub → /docs/meeting-minutes/

D Meeting 4 - Project Charter Finalised

D.1 Date & Attendees

Date: Sunday, 29 March, 2026

Time: 7:30 PM

Location: Online

Chair: Ettore Vescio

Minutes recorded by: Ettore Vescio

Attendees:

- Ettore Vescio (Team Leader)
- Christian Finocchiaro
- Thomas Smith
- Pradip Pandey

D.2 Agenda

1. Finalise Project Charter
2. Plan next steps

D.3 Discussion

D.3.1 Project Charter Finalise

The team finalised the **Project Charter**.

Progress includes:

- All sections reviewed
- Section 3 completed

D.3.2 RSS Feed data

- Viewed RFS feed Data
- Google maps with polygon data working from RFS alert feed

D.4 Decisions Made

- Current team roles remain unchanged
- Assignment 1 Project Charter ready for submission

D.5 Open Questions

- Confirm availability of the Assignment 2 template

D.6 Action Items

Task	Owner	Due
Submit Assignment 1 Project charter	Ettore	Before Due date 30/03/2026
Start wireframe design for front-end	Christian	Before next meeting
Start Data Flow Diagrams for back-end	Thomas	Before next meeting
Start Creating Use cases and user stories	Pradip	Before next meeting
Start ERD design for storage system	Ettore	Before next meeting

D.7 Next Meeting

Date: Sunday, April 12, 2026

Time: 7:30 PM

Minutes stored in: GitHub → /docs/meeting-minutes/



Figure 1: Initial board setup with project sections, workflow columns, and task cards.

Team Name – Critical Signal

Ettore Vescio, Team Leader

Project Manager, Database engineer, Full stack Developer

Thomas Smith, Deputy team leader

Full-stack developer

Christian Finocchiaro

Front-end developer

Pradip Pandey

Security testing, monitoring, and general development support

Figure 2: Team members added, roles assigned, and responsibilities documented on board.

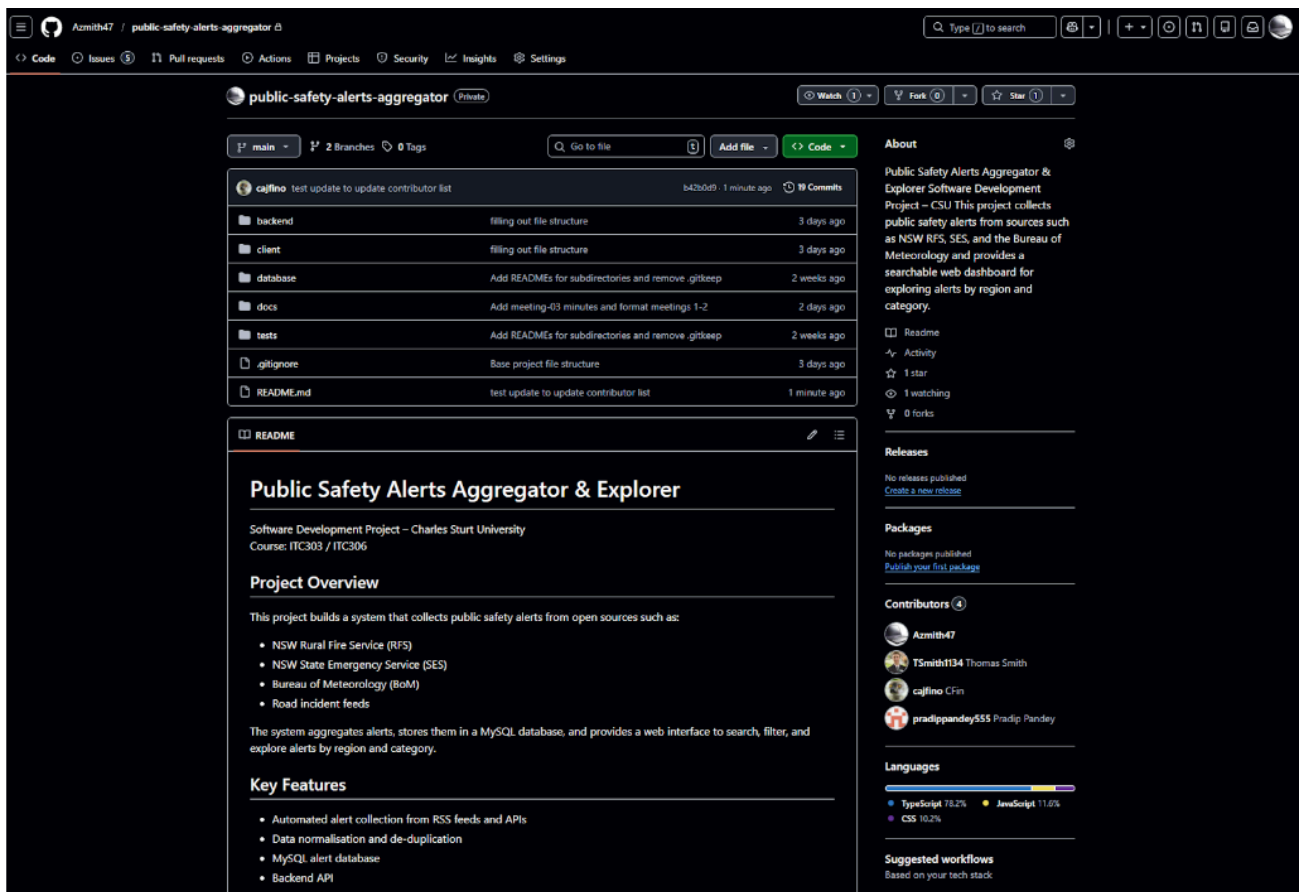


Figure 3: Repository created with README, initial branches, and folder structure.

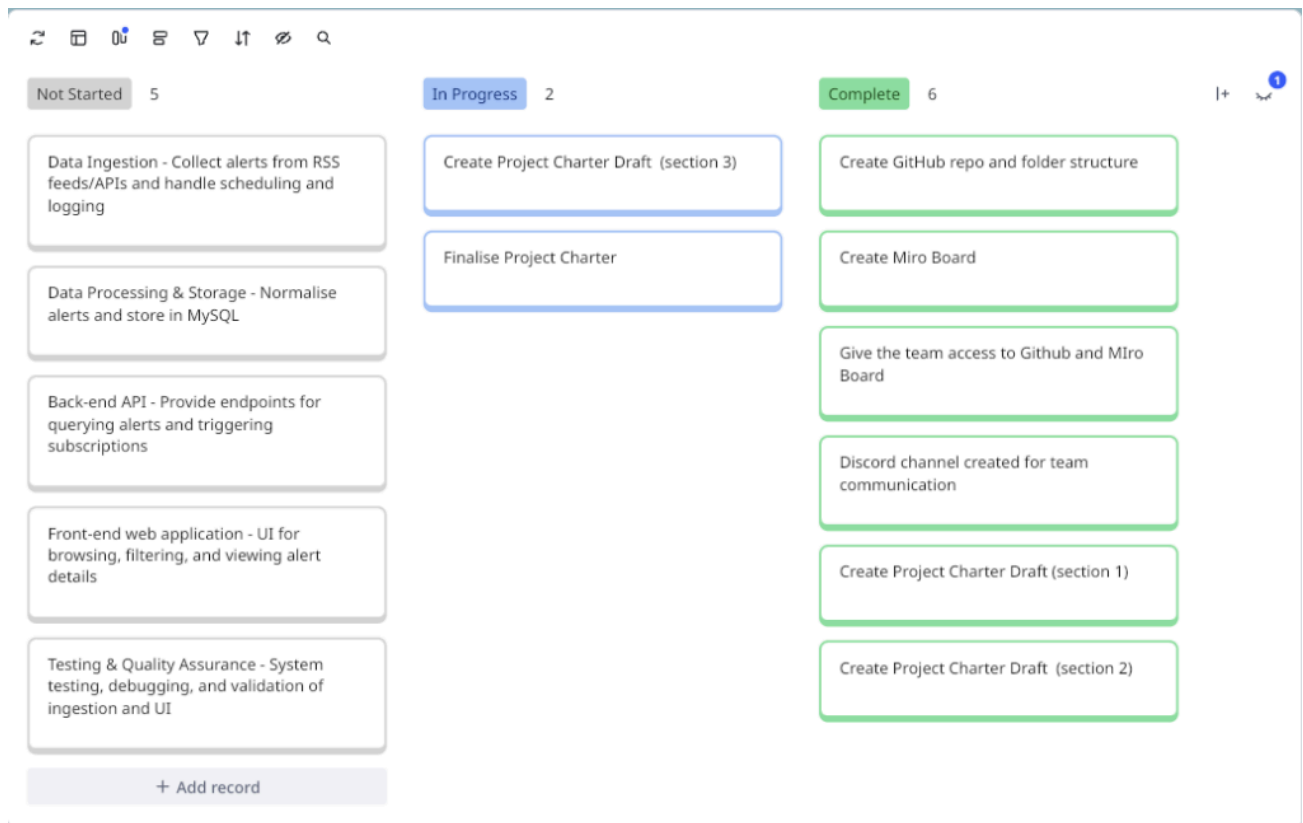


Figure 4: Initial tasks for data ingestion, UI design, and backend API setup logged.

Intro to AI - BD - 1 ...

1 Event

Server Boosts

Game Servers NEW

Information

welcome-and-rules

notes-resources

Text Channels

dev-issues

project-channel

homework-help

session-planning

off-topic

front-end-dev

back-end-dev

database-dev

testing-dev

security-concerns

Voice Channels

Lounge

Study Room 1

Meeting Room 16:59

Christian

pradippandey555

Tom

TV47

Voice Connected

Meeting Room / Intro to AI...

TV47

Meeti

project-channel

Project ideas and discussion

Veterraining

Design Process

Front-end development

Back-end development

Deployment

www.reallygreatsite.com

Tom

3/22/2026 9:22 PM

is software project management, milestones and schedule snapshots serve different, complementary purposes. Milestones are specific, time-duration markers representing significant progress, achievement, or project phases. Schedule snapshots are broad overviews of the entire project timeline at a specific point in time, used to track changes, allocate, and performance over time.

Project Planning Milestones

- Definition: Milestones are key events that signal progress, such as project kick-off, user testing completion, code freeze, or final release.
- Duration: They have zero duration and represent a point in time, not ongoing work.
- Purpose: To define goals, provide direction, maintain momentum, and set tracking points for key stakeholder reviews.
- Software Examples: "Requirements Approved," "Beta Launch Ready," "Code Freeze," "User Acceptance Testing (UAT) Passed."

Schedule Snapshot

- Definition: A snapshot is a time-sensitive, detailed version of a project view or schedule (e.g., Gantt chart).
- Content: It includes the full context – all tasks, dates, dependencies, and resources at that specific moment.
- Purpose: To track project progress (comparing current dates against original deadlines), analyze the impact of delays, and report status history.
- Usage: Often used to compare "Planned" vs. "Actual" progress on a weekly or monthly basis.

Key Differences

Feature	Milestones	Schedule Snapshot
Duration	Zero (a point in time)	None (it spans duration of a task/deliverable)
Content	A single, key achievement	Entire project plan: Tasks, dates, dependencies
Primary Use	Marking key achievement/dependencies	Tracking progress and finding changes
Visibility	High-level (executive-level)	Detailed (project management)
Example	"Alpha Version Complete"	"January 15th Plan vs. Feb 15th Plan"

March 24, 2026

TV47

3/24/2026 7:52 AM

Hey guys I made some changes to the project charter and Miro board. I have been trying to refine wording in each of the sections. I also updated the timeline and added tasks to the kanban board. I will be working on getting section 3 done by Sunday's meeting. Then we can all go through it all and discuss any changes before we submit the assignment

pradippandey555

3/24/2026 8:40 AM

Sure. Thank you

Tom

3/24/2026 1:47 PM

We have recieved emails about our first givernance checkpoint meeting for assessment 3. It will be 11:30am on 1st of April

TV47

3/24/2026 2:27 PM

Yes. Is everyone available next Wednesday for the meeting?

Christian

3/24/2026 2:36 PM

It's a bit tight for me, I have something at 10:30 and usually get home at about 11:30

pradippandey555

3/24/2026 2:37 PM

I can manage for 10-15 mins if needed

Tom

3/24/2026 2:59 PM

The email said it will only be 10-15 mins

+ Message #project-channel

Online — 4

Christian

viewing

pradippandey555

studying

Tom

chilling

TV47

Meeting